

Database Benchmark Report

Automatically generated benchmark report

Date: 2026-07-02 11:39:50

Operating System: Linux

Python: 3.12.3

Databases: H2, HSQLDB

Benchmarks: 8

Measured Runs: 20

Warm-up Runs: 5

Benchmark Summary

Number of databases: **2**

Benchmark categories: **8**

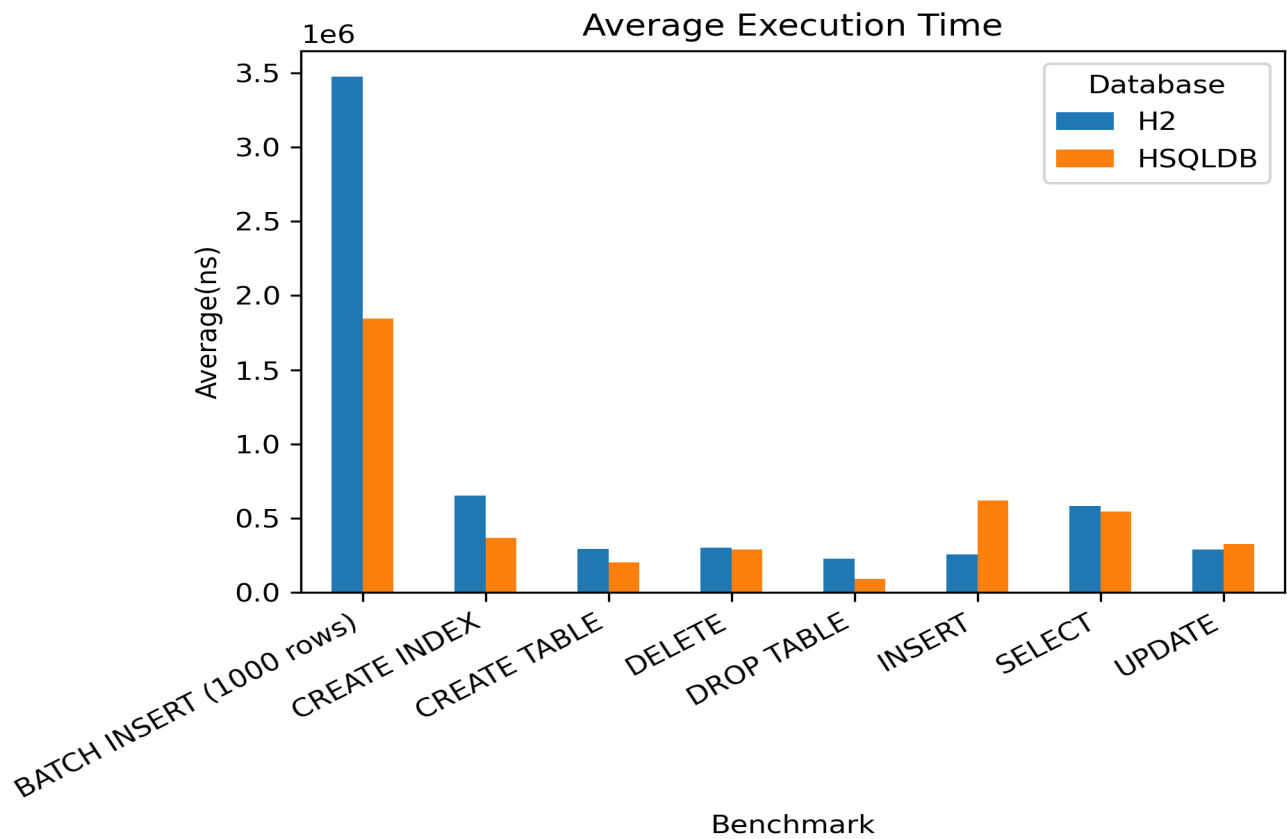
Benchmark results: **16**

This report summarizes execution-time measurements for embedded relational database management systems. All measurements were performed after JVM warm-up iterations and repeated multiple times to improve statistical reliability.

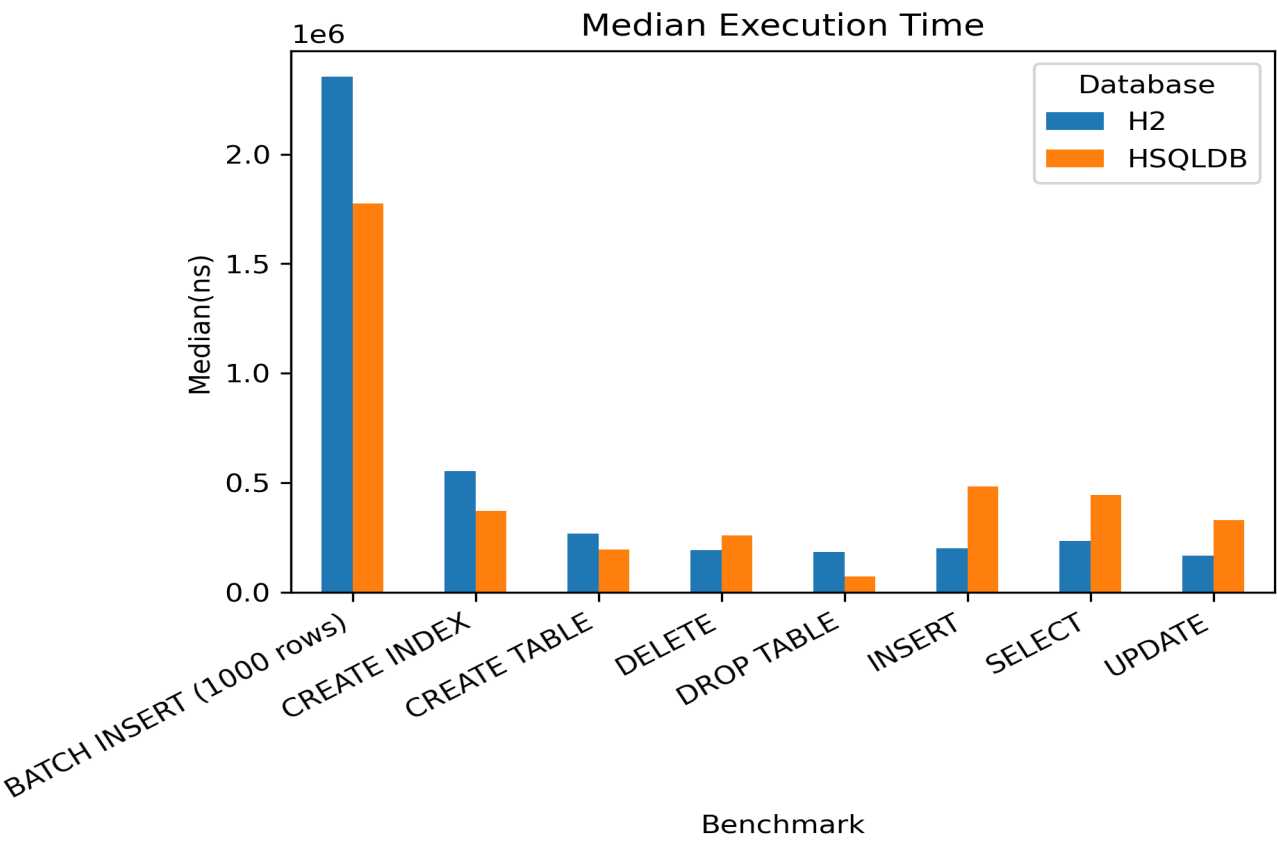
Summary Statistics

Database	Benchmark	Average	Median	Std Dev	Minimum	Maximum
H2	CREATE TABLE	294321	268326	89667	203794	599770
H2	INSERT	256105	200638	172672	131448	905827
H2	BATCH INSERT (1000 rows)	3474332	2353444	2028775	2067766	8589253
H2	SELECT	584075	234266	645152	150854	2737047
H2	UPDATE	288906	169089	289162	120848	1318615
H2	DELETE	303617	192914	362746	109897	1721954
H2	DROP TABLE	227729	184097	135878	116690	585763
H2	CREATE INDEX	652672	552957	231092	444187	1348631
HSQldb	CREATE TABLE	201768	196055	40915	156515	293814
HSQldb	INSERT	620897	484949	492423	235273	2572096
HSQldb	BATCH INSERT (1000 rows)	1846645	1774172	271866	1555902	2801329
HSQldb	SELECT	543126	445980	271091	327618	1496469
HSQldb	UPDATE	328274	329777	48511	244000	415523
HSQldb	DELETE	289940	260371	68834	214834	443015
HSQldb	DROP TABLE	93034	73889	42296	63690	226657
HSQldb	CREATE INDEX	368360	370649	43257	311847	451461

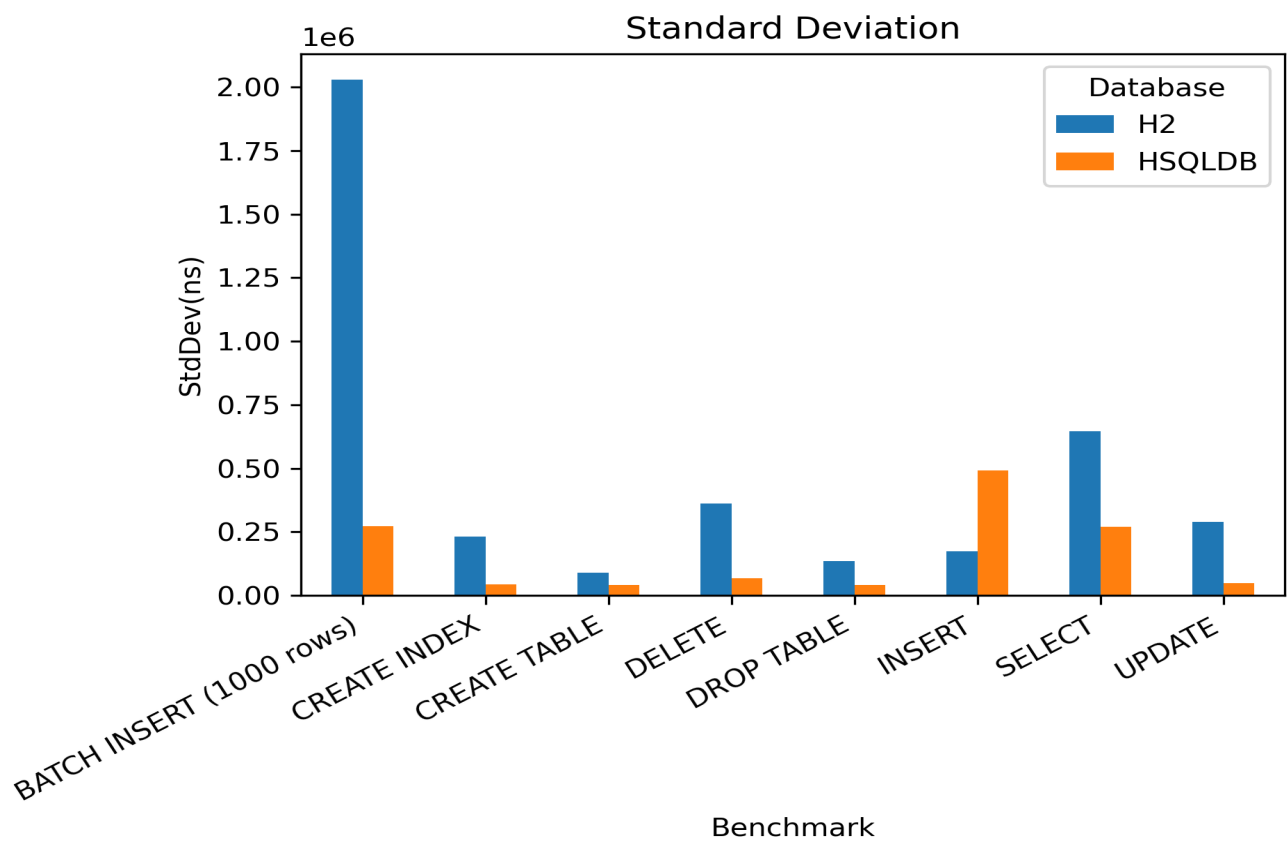
Average Execution Time



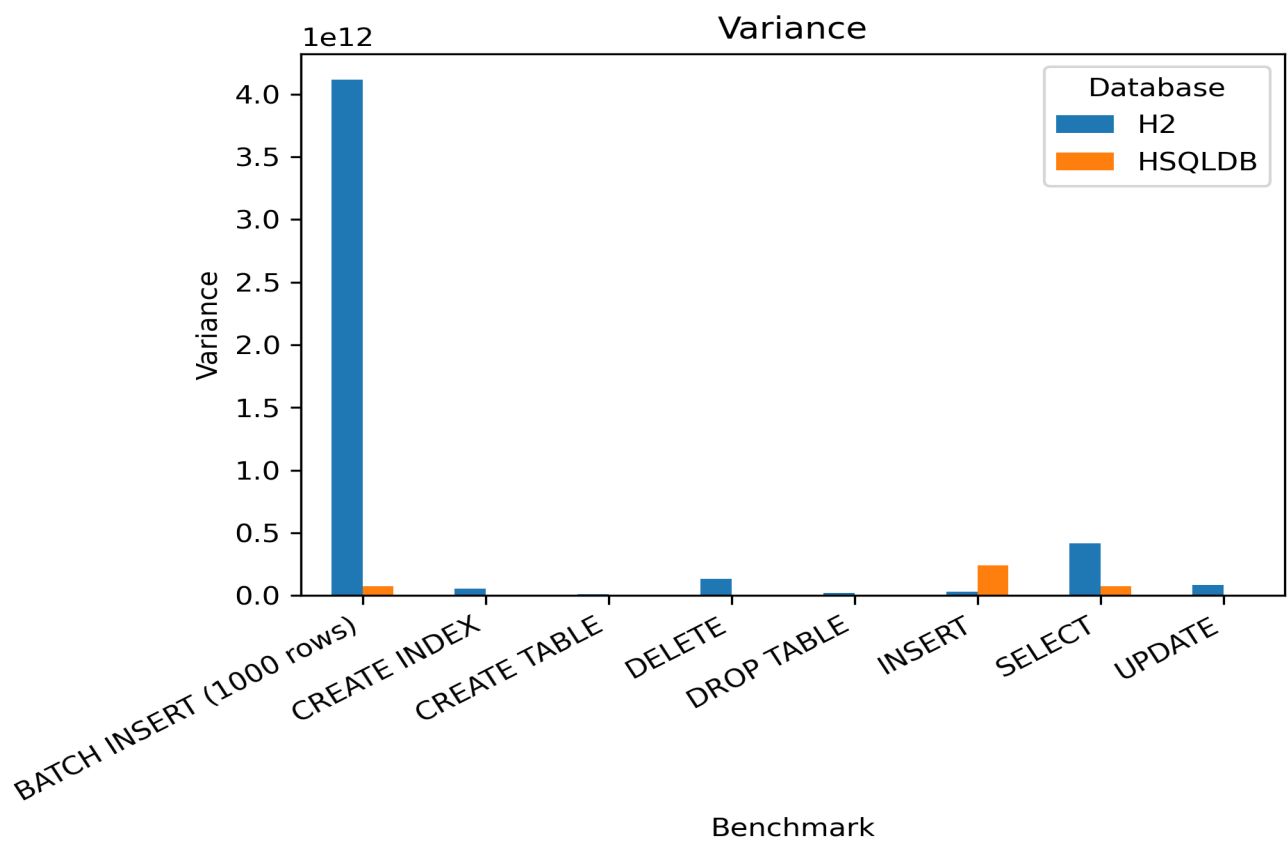
Median Execution Time



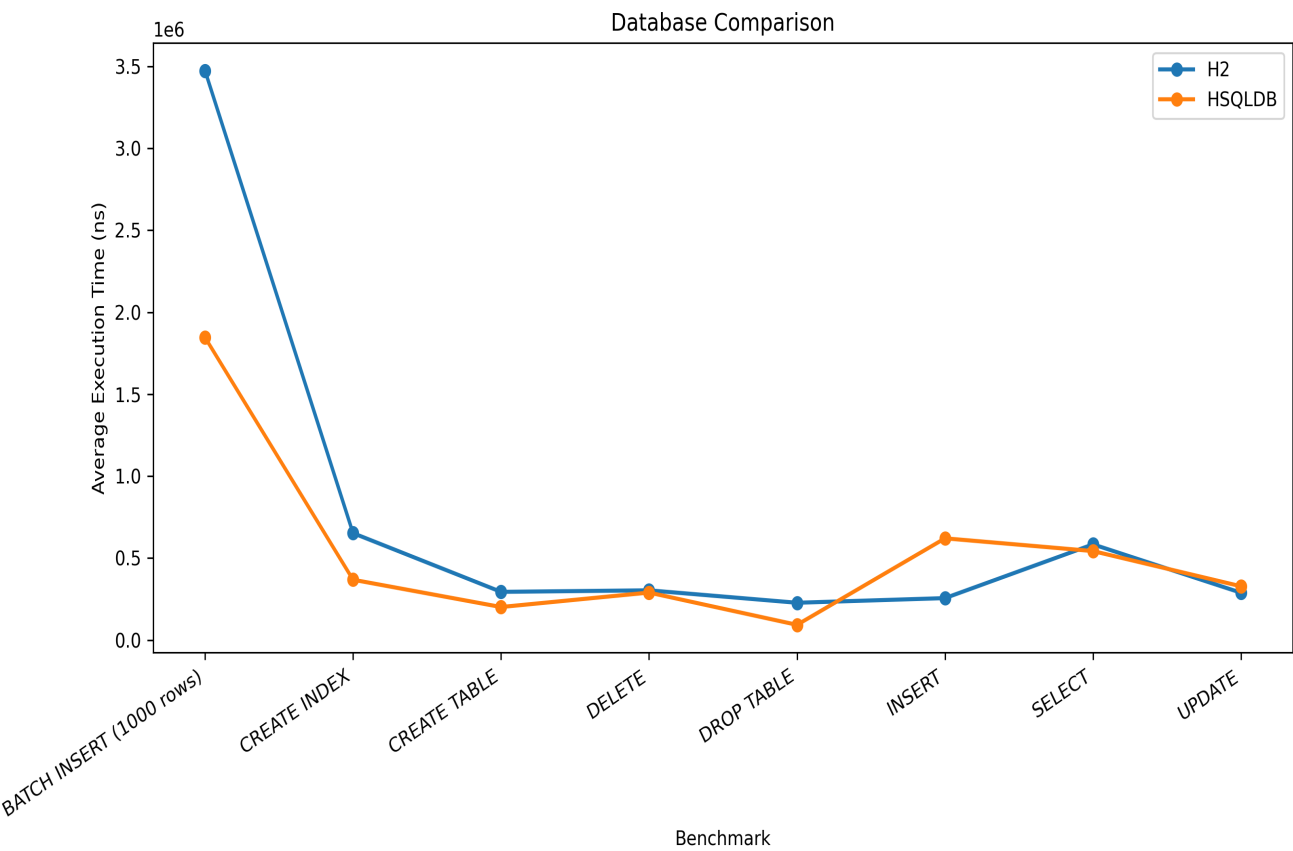
Standard Deviation



Variance



Database Comparison



Benchmark Winners

Benchmark	Winner	Average (ns)
BATCH INSERT (1000 rows)	HSQLDB	1,846,645
CREATE INDEX	HSQLDB	368,360
CREATE TABLE	HSQLDB	201,768
DELETE	HSQLDB	289,940
DROP TABLE	HSQLDB	93,034
INSERT	H2	256,105
SELECT	HSQLDB	543,126
UPDATE	H2	288,906

Overall Benchmark Score

Database	Benchmarks Won
HSQLDB	6
H2	2

Student's t-test

Benchmark	t-statistic	p-value	Significant?
BATCH INSERT (1000 rows)	3.466	0.002484	YES
CREATE INDEX	5.271	0.000035	YES
CREATE TABLE	4.093	0.000354	YES
DELETE	0.161	0.873318	NO
DROP TABLE	4.126	0.000422	YES
INSERT	-3.047	0.005612	YES
SELECT	0.255	0.800717	NO
UPDATE	-0.585	0.564895	NO

Interpretation

A p-value below 0.05 indicates that the observed performance difference between H2 and HSQLDB is statistically significant.

Automatic Conclusion

Summary

The benchmark suite executed 8 benchmark categories on 2 embedded database systems.

Based on the average execution time, **HSQldb** achieved the best overall performance, winning 6 benchmark categories.

The fastest benchmark measured was **DROP TABLE** executed on **HSQldb** with an average execution time of **93,034 ns**.

The slowest benchmark measured was **BATCH INSERT (1000 rows)** executed on **H2** with an average execution time of **3,474,332 ns**.

Overall, the benchmark results indicate that execution times remained stable after warm-up iterations while demonstrating measurable performance differences between the tested database systems.

This report was generated automatically from the benchmark CSV produced by the Java benchmark framework.